Week-5-Assignment

Exercise 1:

Mahendra Singh Dhoni has put up commendable achievements in his baţng career with a boasting batting average of 51.08 (ODIs) & (38.09) in Tests and a striking rate of 89.27 (ODIs) & 59.11 (Tests). In a span of 10-11 years, not only has he become India's most successful captain, but he's also changed the limited-overs game in ways that one could only dream of.

If his entire Test batting statistics are referred, the runs scored by Dhoni in the continual matches followed a particular pattern of series. The series looks like 0, 6, 24, 60, 120, 210, 336, ...

Predict the runs scored by Dhoni in this pattern of series upto "N" matches.

Code:

```
using System;
namespace Exercise_1
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Runs scored by Dhoni in first N matches");
            Console.Write("Enter number of matches played: ");
            string s= Console.ReadLine();
            int.TryParse(s, out int n);
            Console.WriteLine("Runs Scored by Dhoni in "+n+" matches are: \n");
            for(int i=0; i < n; i++)
            {
                int runs = i * (i + 1) * (i + 2);
                Console.Write(runs+" ");
            }
            Console.WriteLine();
        }
    }
}
```

Output:

```
Runs scored by Dhoni in first N matches
Enter number of matches played: 10
Runs Scored by Dhoni in 10 matches are:

0 6 24 60 120 210 336 504 720 990
```

Exercise 2:

Mahi has learnt about the different properties of circles like radius, centre, diameter, circumference ... To make the tutorial class on circles more interesϴng, her teacher organises a game.
The players will be given 2 circles --- One circle A with radius ra and with central coordinate (xa, ya) and second circle B with radius rb and with central coordinate (xb, yb).
The player needs to say
• B is in A
• A is in B
• A and B intersect
• A and B do not intersect
You may assume that A and B are not idenϴcal. Can you please help Mahi in solving this game?

Code:

```
using System;
using System.Runtime.InteropServices;
namespace Exercise_2
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Properties of Circles\n");

            Console.WriteLine("Enter Circle-1 Properties");
            Console.Write("Enter radius of Circle-1: ");
            string sra= Console.ReadLine();
            double.TryParse(sra, out double ra);
            Console.Write("Enter the x-coordinate of Circle-1: ");
            string sxa = Console.ReadLine();
            double.TryParse(sxa, out double xa);
            Console.Write("Enter the y-coordinate of Circle-1: ");
            string sya = Console.ReadLine();
            double.TryParse(sya, out double ya);

            Console.WriteLine("Enter Circle-2 Properties");
            Console.Write("Enter radius of Circle-2: ");
            string srb = Console.ReadLine();
            double.TryParse(srb, out double rb);
            Console.Write("Enter the x-coordinate of Circle-1: ");
            string sxb = Console.ReadLine();
            double.TryParse(sxb, out double xb);
            Console.Write("Enter the y-coordinate of Circle-1: ");
            string syb = Console.ReadLine();
            double.TryParse(syb, out double yb);

            double dis=Math.Sqrt(Math.Pow(xb-xa, 2)+Math.Pow(yb-ya, 2));

            Console.WriteLine();
            if (dis + rb <= ra)
                Console.WriteLine("B is in A");
            else if (dis + ra <= rb)
                Console.WriteLine("A is in B");
            else if (dis <= ra + rb)
                Console.WriteLine("A and B intersect");
            else
                Console.WriteLine("A and B do not intersect");
            Console.WriteLine();
        }
```

```
    }
}
```

Output:

A-is-in-B

```
Properties of Circles

Enter Circle-1 Properties
Enter radius of Circle-1: 4
Enter the x-coordinate of Circle-1: 1
Enter the y-coordinate of Circle-1: 1
Enter Circle-2 Properties
Enter radius of Circle-2: 10
Enter the x-coordinate of Circle-1: 0
Enter the y-coordinate of Circle-1: 0

A is in B
```

B-is-in-A

```
Properties of Circles

Enter Circle-1 Properties
Enter radius of Circle-1: 10
Enter the x-coordinate of Circle-1: 0
Enter the y-coordinate of Circle-1: 0
Enter Circle-2 Properties
Enter radius of Circle-2: 3
Enter the x-coordinate of Circle-1: 2
Enter the y-coordinate of Circle-1: 2

B is in A
```

Intersection

```
Properties of Circles

Enter Circle-1 Properties
Enter radius of Circle-1: 5
Enter the x-coordinate of Circle-1: 0
Enter the y-coordinate of Circle-1: 0
Enter Circle-2 Properties
Enter radius of Circle-2: 5
Enter the x-coordinate of Circle-1: 6
Enter the y-coordinate of Circle-1: 0

A and B intersect
```

No-Intersection

```
Properties of Circles

Enter Circle-1 Properties
Enter radius of Circle-1: 5
Enter the x-coordinate of Circle-1: 0
Enter the y-coordinate of Circle-1: 0
Enter Circle-2 Properties
Enter radius of Circle-2: 5
Enter the x-coordinate of Circle-1: 6
Enter the y-coordinate of Circle-1: 9

A and B do not intersect
```

Exercise: 3

An organization wants to calculate an employee's Net Salary based on:

Basic Salary
HRA = 20% of Basic
DA = 10% of Basic
PF = 12% of Basic

If Basic Salary < 15,000 → No PF deduction. Create a Class Library named SalaryCalculator and create a method in library as signature below:
public static double CalculateNetSalary(double basicSalary)
Consume this library in a console applicaθon.Get proper details of an individual Employee and display proper result.Implement Basic Exception Handling also in Library

Code:

Program.cs

```
using System;
using SalaryCalculator;
namespace Exercise_3
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Net Salary Calculation\n");
            Console.Write("Enter Basic Salary: ");
            string s= Console.ReadLine();
            double.TryParse(s,out double sal);
            double netSal=SalaryCalculation.CalculateNetSalary(sal);
            Console.WriteLine("Net Salary = " + netSal);
        }
    }
}
```

SalaryCalculation.cs

```
namespace SalaryCalculator
{
    public class SalaryCalculation
    {
        public static double CalculateNetSalary(double basicSalary)
        {
            double hra = 0.20 * basicSalary;
            double da = 0.10 * basicSalary;
            double pf = basicSalary < 15000 ? 0 : 0.12 * basicSalary;

            return basicSalary + hra + da - pf;
        }
    }
}
```

Outputs:

Less-Than-15K

```
Net Salary Calculation

Enter Basic Salary: 12000
Net Salary = 15600
```

More-Than-15K

```
Net Salary Calculation

Enter Basic Salary: 80000
Net Salary = 94400
```

Exercise 4:

Create an applicaΘon to produce an Electricity bill using console applicaΘon with proper format. Draw Boxes, lines and columns using characters only.
For input get CustomerId, CustomerName, Address, PhoneNumber, EmailId, Type of connection (Industrial, Business, DomesΘc, Agricultural), Previous Reading, Current Reading.

Calculate Unit Consumed and decide Electricity charges on slab basis as:
First 100 units ₹1.5 / Unit consumed
Next 150 units ₹2.5 / Unit consumed
Next 300 units ₹4.5 / Unit consumed
Above 1000 Units ₹7.5 / Unit consumed
Meter Rent based on Category should be charged as fixed amount.
Industrial 2500
Business 1500
Domestic 1000
Agricultural Free

Code:

```
namespace Exercise_4
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter the following details:\n");
            Console.Write("Customer ID: ");
            string customerId = Console.ReadLine();

            Console.Write("Customer Name: ");
            string customerName = Console.ReadLine();

            Console.Write("Address: ");
            string address = Console.ReadLine();

            Console.Write("Phone Number: ");
            string phone = Console.ReadLine();

            Console.Write("Email ID: ");
            string email = Console.ReadLine();

            Console.Write("Connection Type 
(Industrial/Business/Domestic/Agricultural): ");
            string connectionType = Console.ReadLine();

            Console.Write("Previous Reading: ");
            string s=Console.ReadLine();
            int.TryParse(s,out int prevReading);

            Console.Write("Current Reading: ");
            string t=Console.ReadLine();
            int.TryParse(t, out int currReading);

            int unitsConsumed = currReading - prevReading;
            double energyCharge = CalculateBill(unitsConsumed);
            double meterRent = GetMeterRent(connectionType);
            double totalAmount = energyCharge + meterRent;


Console.WriteLine("┌──────────────────────────────────────────────────────────┐")
;
```

```
            Console.WriteLine("│                    ELECTRICITY BILL
│");
Console.WriteLine("├────────────────────────────────────────────────────────────┤")
;
            Console.WriteLine($"│ Customer ID        : {customerId,-37}│");
            Console.WriteLine($"│ Customer Name      : {customerName,-37}│");
            Console.WriteLine($"│ Address            : {address,-37}│");
            Console.WriteLine($"│ Phone Number       : {phone,-37}│");
            Console.WriteLine($"│ Email ID           : {email,-37}│");
            Console.WriteLine($"│ Connection Type    : {connectionType,-37}│");
Console.WriteLine("├────────────────────────────────────────────────────────────┤")
;
            Console.WriteLine($"│ Previous Reading   : {prevReading,-37}│");
            Console.WriteLine($"│ Current Reading    : {currReading,-37}│");
            Console.WriteLine($"│ Units Consumed     : {unitsConsumed,-37}│");
Console.WriteLine("├────────────────────────────────────────────────────────────┤")
;
            Console.WriteLine($"│ Energy Charges     : Rs. {energyCharge,-33}│");
            Console.WriteLine($"│ Meter Rent         : Rs. {meterRent,-33}│");
Console.WriteLine("├────────────────────────────────────────────────────────────┤")
;
            Console.WriteLine($"│ TOTAL AMOUNT       : Rs. {totalAmount,-33}│");
Console.WriteLine("└────────────────────────────────────────────────────────────┘")
;
            Console.WriteLine();
        }
        static double CalculateBill(int units)
        {
            double amount = 0;


             if (units <= 100)
                amount = units * 1.5;
            else if (units <= 250)
                amount = (100 * 1.5) + ((units - 100) * 2.5);
            else if (units <= 550)
                amount = (100 * 1.5) + (150 * 2.5) + ((units - 250) * 4.5);
            else
                amount = (100 * 1.5) + (150 * 2.5) + (300 * 4.5) + ((units - 550)
* 7.5);

            return amount;
        }

        static double GetMeterRent(string type)
        {
            switch (type.ToLower())
            {
                case "industrial":
                    return 2500;
                case "business":
                    return 1500;
                case "domestic":
                    return 1000;
                case "agricultural":
                    return 0;
                default:
                    return 0;
```

```
                }
            }
        }
}
```

Outputs:

Agriculture

```
Enter the following details:

Customer ID: 7
Customer Name: MS Dhoni
Address: Ranchi
Phone Number: 9121059578
Email ID: thala@gmail.com
Connection Type (Industrial/Business/Domestic/Agricultural): agriculture
Previous Reading: 7092
Current Reading: 7800
┌──────────────────────────────────────────────────────────┐
│                    ELECTRICITY BILL                      │
├──────────────────────────────────────────────────────────┤
│ Customer ID       : 7                                    │
│ Customer Name     : MS Dhoni                             │
│ Address           : Ranchi                               │
│ Phone Number      : 9121059578                           │
│ Email ID          : thala@gmail.com                      │
│ Connection Type   : agriculture                          │
├──────────────────────────────────────────────────────────┤
│ Previous Reading  : 7092                                 │
│ Current Reading   : 7800                                 │
│ Units Consumed    : 708                                  │
├──────────────────────────────────────────────────────────┤
│ Energy Charges    : Rs. 3060                             │
│ Meter Rent        : Rs. 0                                │
├──────────────────────────────────────────────────────────┤
│ TOTAL AMOUNT      : Rs. 3060                             │
└──────────────────────────────────────────────────────────┘
```

Business

```
Enter the following details:

Customer ID: 45
Customer Name: Rohit Sharma
Address: Mumbai
Phone Number: 9121059392
Email ID: rohit.sharma@gmail.com
Connection Type (Industrial/Business/Domestic/Agricultural): business
Previous Reading: 26498
Current Reading: 28000
┌──────────────────────────────────────────────────────────┐
│                    ELECTRICITY BILL                      │
├──────────────────────────────────────────────────────────┤
│ Customer ID       : 45                                   │
│ Customer Name     : Rohit Sharma                         │
│ Address           : Mumbai                               │
│ Phone Number      : 9121059392                           │
│ Email ID          : rohit.sharma@gmail.com               │
│ Connection Type   : business                             │
├──────────────────────────────────────────────────────────┤
│ Previous Reading  : 26498                                │
│ Current Reading   : 28000                                │
│ Units Consumed    : 1502                                 │
├──────────────────────────────────────────────────────────┤
│ Energy Charges    : Rs. 9015                             │
│ Meter Rent        : Rs. 1500                             │
├──────────────────────────────────────────────────────────┤
│ TOTAL AMOUNT      : Rs. 10515                            │
└──────────────────────────────────────────────────────────┘
```

Domestic

```
Enter the following details:

Customer ID: 96
Customer Name: Shreyas Iyer
Address: Mumbai
Phone Number: 9121078345
Email ID: shreyas@gmail.com
Connection Type (Industrial/Business/Domestic/Agricultural): domestic
Previous Reading: 9623
Current Reading: 9800
+------------------------------------------------------------+
|                     ELECTRICITY BILL                       |
+------------------------------------------------------------+
| Customer ID        : 96                                    |
| Customer Name      : Shreyas Iyer                          |
| Address            : Mumbai                                |
| Phone Number       : 9121078345                            |
| Email ID           : shreyas@gmail.com                     |
| Connection Type    : domestic                              |
+------------------------------------------------------------+
| Previous Reading   : 9623                                  |
| Current Reading    : 9800                                  |
| Units Consumed     : 177                                   |
+------------------------------------------------------------+
| Energy Charges     : Rs. 342.5                             |
| Meter Rent         : Rs. 1000                              |
+------------------------------------------------------------+
| TOTAL AMOUNT       : Rs. 1342.5                            |
+------------------------------------------------------------+
```

Industrial

```
Enter the following details:

Customer ID: 18
Customer Name: Virat Kohli
Address: Delhi
Phone Number: 9121057380
Email ID: kohli@gmail.com
Connection Type (Industrial/Business/Domestic/Agricultural): industrial
Previous Reading: 83009
Current Reading: 89000
+------------------------------------------------------------+
|                     ELECTRICITY BILL                       |
+------------------------------------------------------------+
| Customer ID        : 18                                    |
| Customer Name      : Virat Kohli                           |
| Address            : Delhi                                 |
| Phone Number       : 9121057380                            |
| Email ID           : kohli@gmail.com                       |
| Connection Type    : industrial                            |
+------------------------------------------------------------+
| Previous Reading   : 83009                                 |
| Current Reading    : 89000                                 |
| Units Consumed     : 5991                                  |
+------------------------------------------------------------+
| Energy Charges     : Rs. 42682.5                           |
| Meter Rent         : Rs. 2500                              |
+------------------------------------------------------------+
| TOTAL AMOUNT       : Rs. 45182.5                           |
+------------------------------------------------------------+
```

Exercise-5:

Boxing is a martial art and combat sport in which two people throw punches at each other, usually
with gloved hands. Historically, the goals have been to weaken and knock down the opponent. In
December 2025, the World Boxing Council, World Boxing AssociaΘon and the InternaΘonal Boxing
FederaΘon reached an agreement to standardize the names for January 2026 onwards, of the different
weight categories under which Boxing matches are conducted.

John is a good boxer and he wants to know under what category he should enrol his name.
Can you please help him out?

Input Format :
Input consists of an integer that corresponds to John's weight.

Output Format :
Output consists of a string that corresponds to any of the category names listed above or “Invalid Input”.
Print Invalid Input if the weight is negative or if the weight is greater than 120.

Code:

```
namespace Exercise_5
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("John's Weight Classification\n");
            Console.Write("Enter John's Weight: ");
            string s=Console.ReadLine();
            int.TryParse(s, out int weight);

            if (weight < 0 || weight > 120)
            {
                Console.WriteLine("Invalid Input");
            }
            else if (weight <= 48)
            {
                Console.WriteLine("Light Fly");
            }
            else if (weight <= 51)
            {
                Console.WriteLine("Fly");
            }
            else if (weight <= 54)
            {
                Console.WriteLine("Bantam");
            }
            else if (weight <= 57)
            {
                Console.WriteLine("Feather");
            }
            else if (weight <= 60)
            {
                Console.WriteLine("Light");
```

```
            }
            else if (weight <= 64)
            {
                Console.WriteLine("Light Welter");
            }
            else if (weight <= 69)
            {
                Console.WriteLine("Welter");
            }
            else if (weight <= 75)
            {
                Console.WriteLine("Light Middle");
            }
            else if (weight <= 81)
            {
                Console.WriteLine("Middle");
            }
            else if (weight <= 91)
            {
                Console.WriteLine("Light Heavy");
            }
            else
            {
                Console.WriteLine("Heavy");
            }
        }
    }
}
```

<u>Output:</u>